HA panel main board — schematic index
Sheet 2: 20 V inlet, fuse, TPS16630 eFuse, bus TVS
Sheet 3: NID100-5 5 V supply, Pi USB-C output, radar 5 V
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Sheet 5: TAS5756M amplifier and speaker outputs

20 V inlet and eFuse



File: inlet.kicad_sch

5 V supply and outputs



File: supply_5v.kicad_sch

3.3 V, Pi link, sensors, radar, echo-ref



File: io.kicad_sch

Amplifier



File: amplifier.kicad_sch

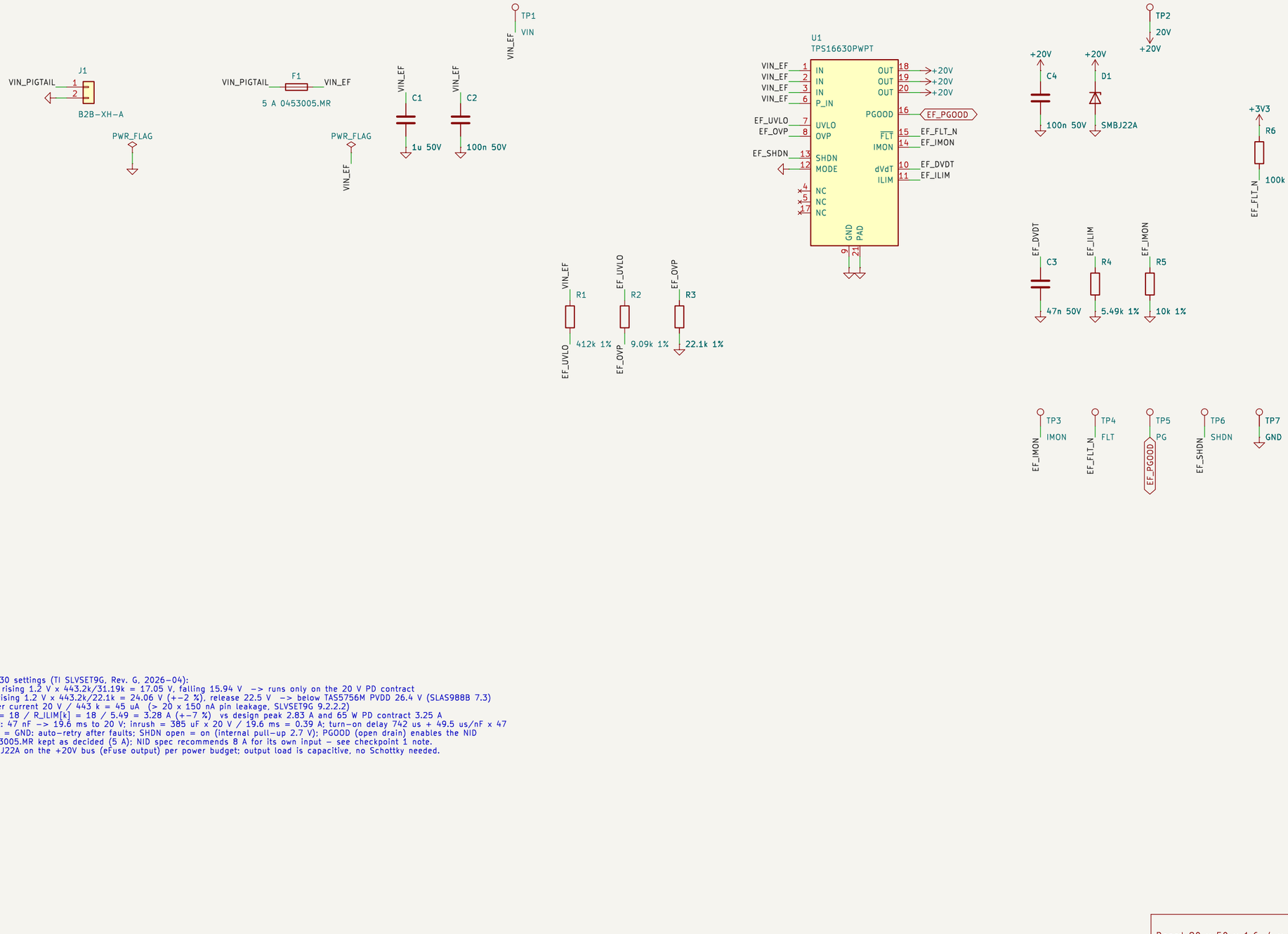
Board 90 x 50 x 1.6, 4 x Ø2.7 on 83 x 43
Brief claude/brief-electronics-design-2026-11-07.md
KiCad 9.0.9 — generated by hardware/main-board/gen (checkpoint 1)
HA panel (neocleous/ha-panel)

Sheet: /
File: main-board.kicad_sch

Title: HA panel main board

Size: A4	Date: 2026-11-07	**Rev: 1**
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20 V INLET (5807 pigtail -> fuse -> TPS16630 eFuse -> +20V bus)



TPS16630 settings (TI SLVSET9G, Rev. G, 2026-04):
  UVLO rising 1.2 V x 443.2k/31.19k = 17.05 V, falling 15.94 V -> runs only on the 20 V PD contract
  OVP rising 1.2 V x 443.2k/22.1k = 24.06 V (+-2 %), release 22.5 V -> below TAS5756M PVDD 26.4 V (SLAS988B 7.3)
  Divider current 20 V / 443 k = 45 uA (> 20 x 150 nA pin leakage, SLVSET9G 9.2.2.2)
  I_OL = 18 / R_ILIM[k] = 18 / 5.49 = 3.28 A (+-7 %) vs design peak 2.83 A and 65 W PD contract 3.25 A
  dV/dt: 47 nF -> 19.6 ms to 20 V; inrush = 385 uF x 20 V / 19.6 ms = 0.39 A; turn-on delay 742 us + 49.5 us/nF x 47
  MODE = GND: auto-retry after faults; SHDN open = on (internal pull-up 2.7 V); PGOOD (open drain) enables the NID
F1 0453005.MR kept as decided (5 A); NID spec recommends 8 A for its own input - see checkpoint 1 note.
D1 SMBJ22A on the +20V bus (eFuse output) per power budget; output load is capacitive, no Schottky needed.

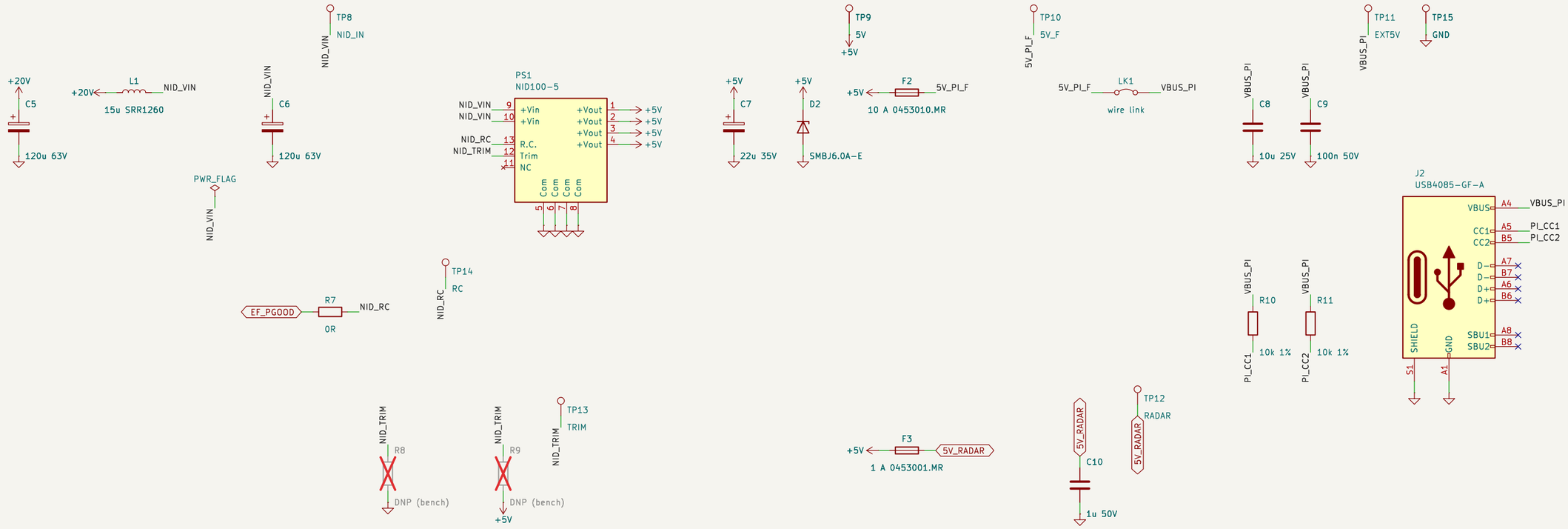
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Sheet: /20 V inlet and eFuse/
File: inlet.kicad_sch

Title: Main board - 20 V inlet and eFuse

Size: A3	Date: 2026-11-07	**Rev: 1**
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5 V SUPPLY (Mean Well EMC circuit -> NID100-5 -> Pi USB-C output, radar branch)



Mean Well NID100-SPEC (2024-10-30): EMC circuit C1/C2 120 uF/63 V, L1 15 uH (NiZn), C3 22 uF/35 V; ext. C max 100 uF/16 V;
  fuse recommended 8 A; R.C.: on = open or 1.2-12 V, off = < 0.4 V; pin 12 "Trim (optional)", no formula published.
L1 is a single winding in +Vin: the input return is the common board ground (non-isolated NID, amp on the same ground).
External C on +5V: C7 22 uF + C8 10 uF (X7R 1206, -70 % at 5 V) + C9 + C10 + LDO input path (behind 2.2 Ohm) - 35 uF << 100 uF.
PI output: 0453010.MR -> LK1 (wire link) -> J2 USB4085 VBUS; CC1/CC2 Rp 10 k to 5 V (3 A); Pi uses PSU_MAX_CURRENT=5000.
J2 USB4085-GF-A (THT, decision 3A) sits on the door side, mouth down at the bottom edge; all other parts on the Pi side.
Bench 2.7 test pads: 5V (NID out) | 5V_F (after fuse) | EXT5V (at J2) ; LK1 opened for the series current.
Trim network DNP (D23/V20): R8 Trim->Com raises Vout, R9 Trim->+Vout lowers it (FB-divider convention, not published
  by Mean Well). Spec lists pin 12 as "Trim (optional)" with no formula; on arrival measure R(pin 12-Com) unpowered and
  Vout with a trimmer clipped from TP13 to GND before fitting R8. No response at pin 12 -> alternative-regulator decision.

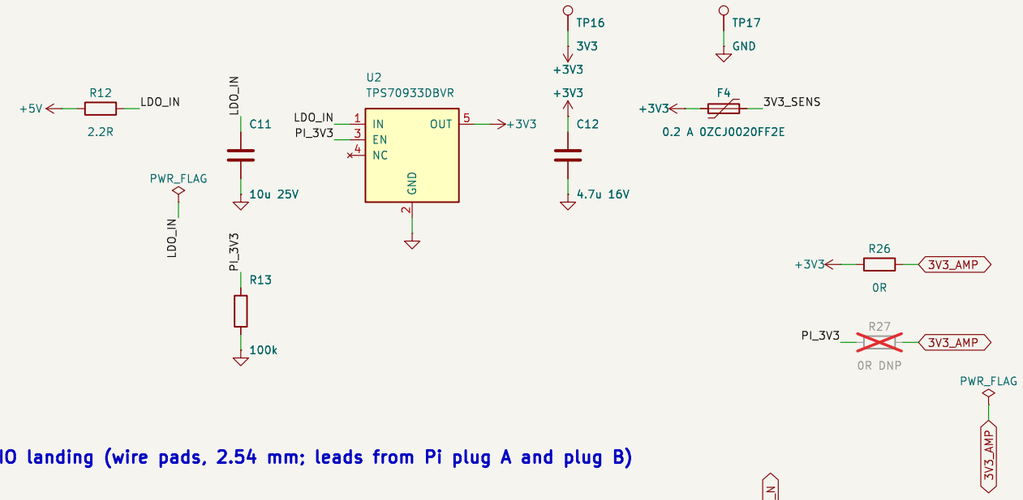
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Sheet: /5 V supply and outputs/
File: supply_5v.kicad_sch

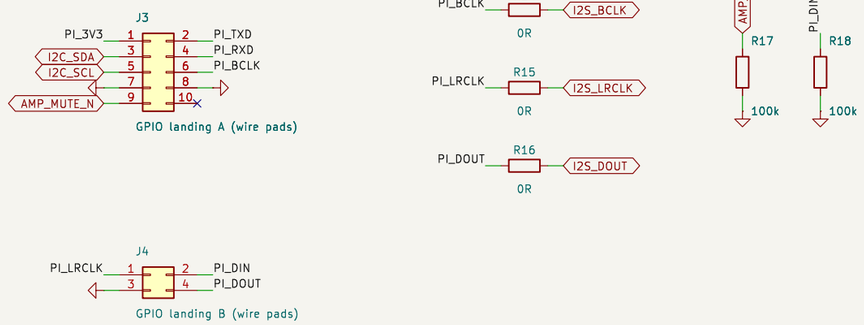
Title: Main board — 5 V supply and outputs

Size: A3	Date: 2026-11-07	**Rev: 1**
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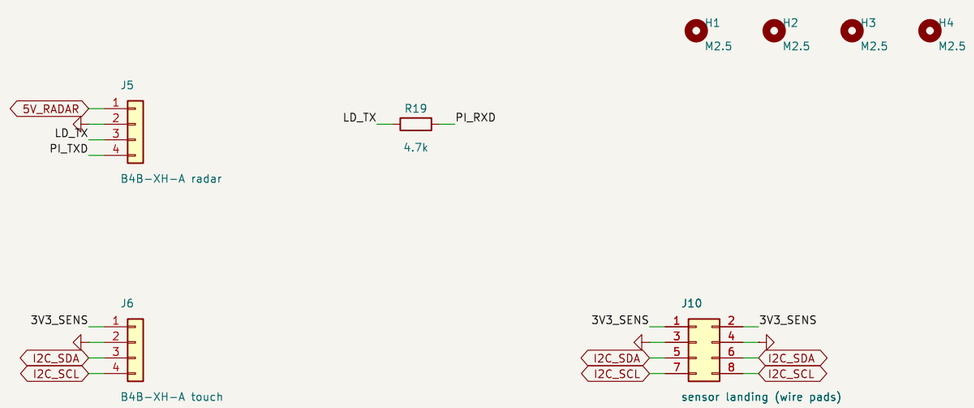
3.3 V LDO (follows the Pi 3V3)



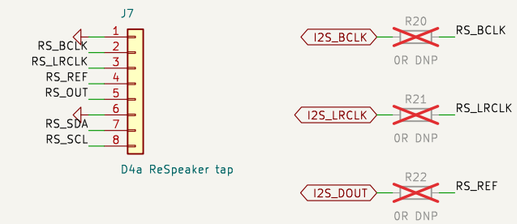
Pi GPIO landing (wire pads, 2.54 mm; leads from Pi plug A and plug B)



XH headers (JST XH 2.50 mm) – radar, touch board



Echo-reference (D4a) – default topology A: all links OPEN



V17: Raspberry Pi publishes no current rating for the Pi 5 header 3V3 pins -> 3.3 V LDO on this board (brief item 4).
U2 TPS70933: 2.7–30 V in (survives the NID OVP 6.4–7.5 V). 150 mA, EN follows the Pi 3V3 (pin 1 lead, ~33 uA).
Loads: TAS5756M DVDD/AVDD/CPVDD 35 mA (SLAS988B 7.12) + touch/BME680/VEML6030 via F4.
STANDBY (Pi 3V3 off, 5 V on, HAT+ spec 2.2): LDO off -> no back-feed through I2C; amp muted by R17.
R26/R27: amp logic rail 3V3_AMP from the LDO (default, R26) or from the Pi 3V3 (R27): TAS5756M VIH = 0.9 x DVDD (SLAS988B 7.3) and TI advises driving its inputs from the DVDD supply; LDO +-1 % (SBVS186H) vs Pi 3V3 (not published).
Pi pins: 3/5 SDA/SCL (GPIO2/3, 1.8 k pull-ups on the Pi); 8/10 TXD/RXD (GPIO14/15, uart0):
12/35/38/40 = GPIO18/19/20/21 I2S BCLK/LRCLK/DIN/DOUT; 15 = GPIO22 amp mute (rpi-digiampplus).
I2C map: AT42QT1070 0x1B, VEML6030 0x48, TAS5756M 0x4C, BME680 0x76/0x77 (ReSpeaker 0x42/0x18 only if R24/R25 fitted).
Sensor branch 3V3_SENS (F4 0.2 A PTC): J6 touch board, J10 BME680 + VEML6030 (leads moved from the touch board, 5A).
Topology B (bench 2.2): fit R20–R23 (+R24/R25 for XU316 I2C), jumper J7 to the ReSpeaker XIAO footprint, I2S firmware.

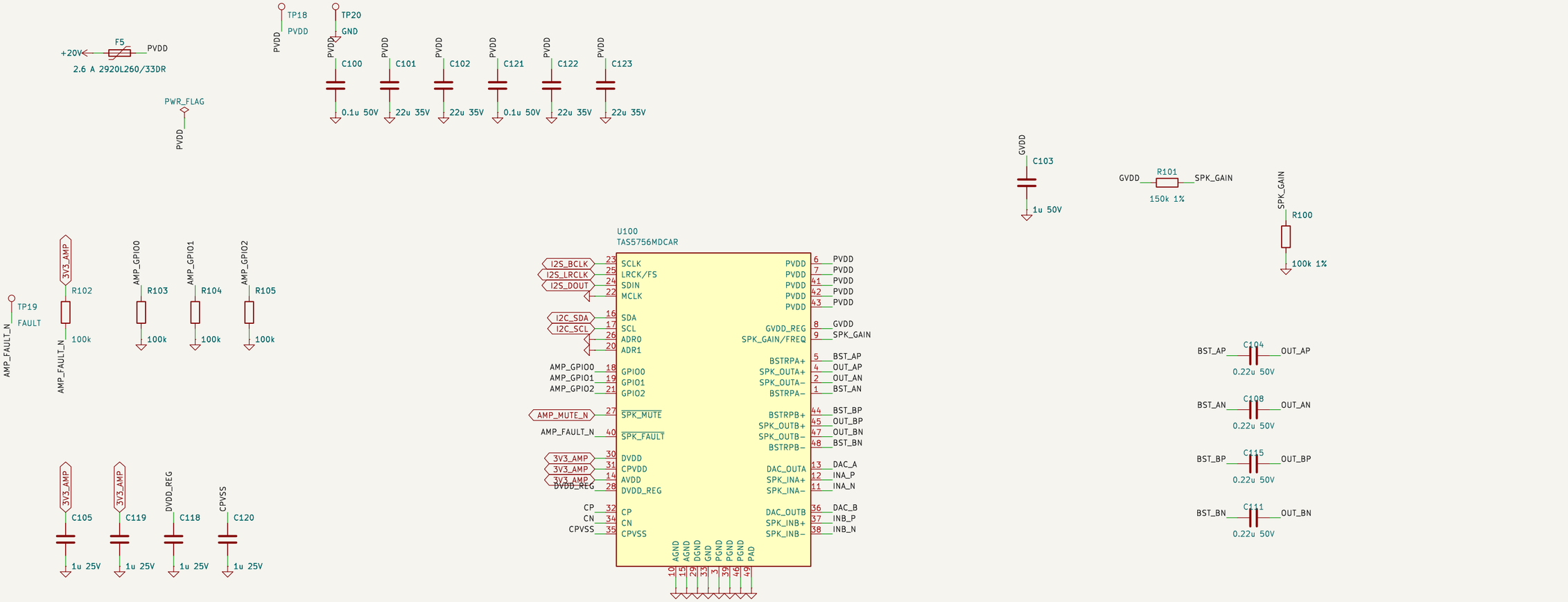
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Sheet: /3.3 V, Pi link, sensors, radar, echo-ref/
File: io.kicad_sch

Title: Main board – 3.3 V, Pi link, XH, D4a

Size: A3	Date: 2026-11-07	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 4/5

TAS5756M stereo BTL (TI SLAS988B Fig. 79 / Table 22; DigiAMP+ wiring for the rpi-digiampplus overlay)



Gain/switching: SPK_GAIN = GVDD_REG x 100k/250k = 2.8 V at GVDD 7.0 V -> 20 dBV, 8 x fSYNC = 768 kHz @48 kHz (Table 15, window 2.33–3.10 V holds for GVDD 5.83–7.75 V). D22: software ceiling −4.0 dBFS keeps 10 W into 4 Ohm.
LC: type 2 (L per leg, C to GND) as Fig. 79 and SLAA701B 1.3.2 for BD modulation; 10 uH + 0.68 uF: f0 61 kHz, Q 0.52 at 4 Ohm (−0.8 dB at 20 kHz). Power budget cited SLAA701B 1.3.4.1 = type 1 (differential C) − corrected.
CSBA0B50-100M: Isat 3.50 A (L −10 %) vs normal peak 2.6 A (10 W/4 Ohm 2.24 A + 0.33 A ripple at 768 kHz); at the 7.5 A OCE L >= 7 uH (−30 % at 8.00 A) vs 4.7 uH nominal / 1 uH min under short circuit (SLAS988B 7.3; CODACA 2026-03-16).
Caps: TI Table 22 values; 0402 parts moved to 0603 (hand assembly D42). PVDD caps > 1.45 x PVDD, LC caps > 1.8 x PVDD.
Overlay rpi-digiampplus: pcm5122@4c on i2c1, i2s_clk_producer, mute-gpios GPIO22 active high (driver sets 1 = unmute).
Thermal pad: via array + paste windows 70–80 % of the pad area (SLAS988B 11.1.3.2) − footprint in block 2.
No PVDD electrolytic at the amp (block 2, layout density): TI Table 22 says the 390 uF bulk "can be reduced or even removed"; C5 (120 uF on +20V) feeds PVDD through F5 (75 mOhm max, tau 9 us), plus 4 x 22 uF + 2 x 0.1 uF at the PVDD pins.

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Sheet: /Amplifier/
File: amplifier.kicad_sch

Title: Main board − amplifier

Size: A3	Date: 2026-11-07	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 5/5